EMBER SAFE | SMARTLOOP RANGE

SmartLoop-4

Installation, Commissioning & Maintenance Manual

INSTALLATION, COMMISSIONING & MAINTENANCE



FOR TRAINED STAFF **Installation, commissioning, maintenance and troubleshooting of the SmartLoop-4 system.**

INFUSIVE INDOTECH PRIVATE LIMITED

Ember Safe | Stay Alert

Document ES-SL4-IM Revision 1.0 November 2026

Important Notices

About this manual

This manual is for the installers, commissioning engineers and maintenance staff of the Ember Safe SmartLoop-4 analogue addressable fire detection and alarm system.

It covers system design rules, mounting, wiring, commissioning, maintenance, troubleshooting, check lists and the Do's & Don'ts card, as required by RDSO/SPN/217/2025 Version 3.1 cl. 6.9 and 11.1. Operation of the panel is in the Operation Manual (ES-SL4-OM).

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Disclaimer

The information in this manual is believed to be accurate at the time of publication. Infusive Indotech Private Limited reserves the right to change product features and this document without notice as part of continuous product improvement. Illustrations are representative; the appearance of your product may differ. A fire alarm system is a life-safety system: its design, installation, commissioning and maintenance must be carried out by trained, competent persons in line with the fire codes and the requirements of the local authority having jurisdiction.

Symbols used in this manual

Symbol	Meaning
WARNING	Risk to life or of serious injury, or a fire alarm that may not work. Always follow.
CAUTION	Risk of damage to the equipment or of false alarms.
NOTE	Useful information for correct operation.
GOOD PRACTICE	Recommendation from Ember Safe for the best results.

Revision history

Revision	Date	Description
1.0	November 2026	First edition.

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1 Safety

WARNING Mains voltage (110 / 230 V AC) is present inside the panel. Switch off and isolate the mains and disconnect the battery before opening, wiring or changing cards.

The system protects life and railway assets. Before any work that can start an alarm, suppression or power shut-down, inform the Station Master and disable the affected outputs. Enable them again when the work is finished.

CAUTION Cards are static sensitive: wear an earthed wrist strap and hold cards by their edges.

Use only Ember Safe devices and spares approved for the SmartLoop-4.

Lead-acid batteries hold a large charge: do not short the terminals; fit the battery last.

NOTE Work to IS 2189, the Indian Electricity Rules / CEA regulations, RDSO/SPN/217/2025 Version 3.1 and the Zonal Railway's instructions.

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2 System Description

System overview

The SmartLoop-4 is an analogue addressable fire detection and alarm system for Signalling & Telecom installations. One control panel serves 4 loops of up to 128 addressable devices each. Every device has a 32-bit serial number and a loop address (001 to 128). The panel polls every device, reads its analogue value and state, and acts on the action level programmed for it.

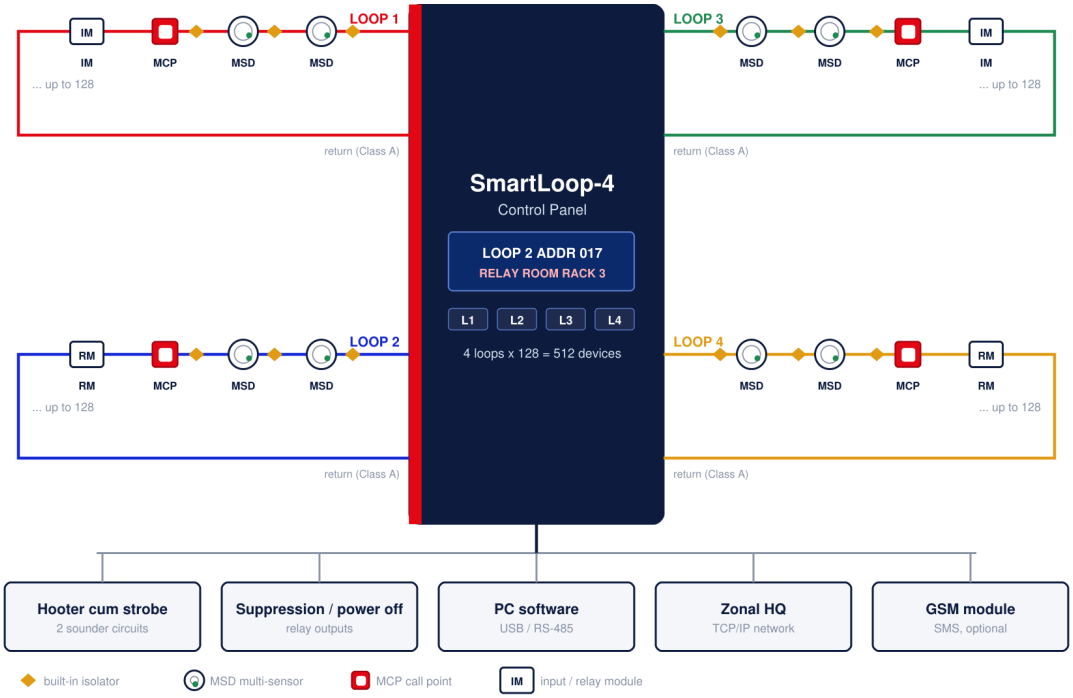


Figure 1 System overview

Panel modules

The panel is modular (RDSO/SPN/217/2025 Version 3.1 cl. 2.2): a faulty card is replaced by a spare without changing the rest.

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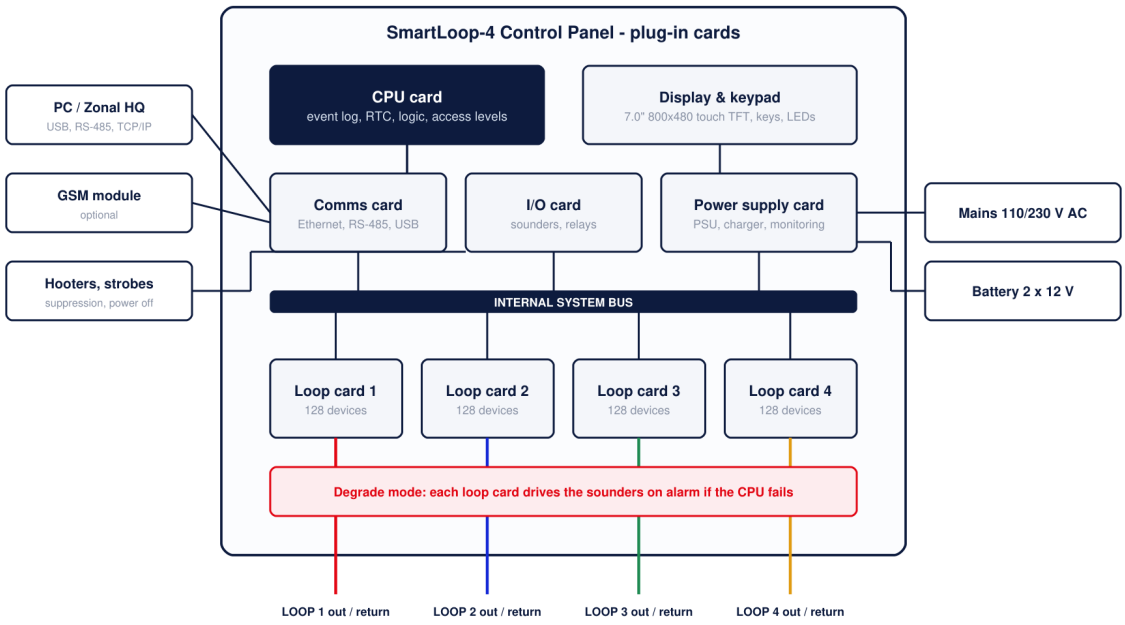


Figure 2 Block diagram - mounting arrangement of the cards (cl. 11.1.2.1)

Table 1 **Panel cards**

Card	Function
CPU card	System logic, action levels, coincidence pairs, event log (512 + 512), real-time clock, access levels, configuration memory.
Loop card (x4)	Drives one loop: 24 V loop supply, two-way communication with 128 devices, open / short / earth supervision, Class A return. Degrade mode: sounds the alarm on its own if the CPU fails.
Display & keypad	7.0-inch 800 x 480 colour TFT smart display with capacitive touch (DGUS II, DMG80480C070_04WTC) on a serial link to the CPU; RESET, SILENCE, OK, arrow keys, key switch, status and loop LEDs, buzzer.
I/O card	2 monitored sounder circuits, FIRE and FAULT relays, 3 NO + 3 NC programmable relays, aux 24 V.
Comms card	Ethernet (TCP/IP), RS-485, USB; GSM module port.
Power supply card	Mains PSU 110 / 230 V AC, 24 V DC, battery charger, mains / battery / charger / earth monitoring, reverse polarity protection.

Loop communication

The loop carries 24 V DC power and digital communication on the same two wires. The panel sends commands by short changes of the loop voltage; devices answer by drawing current pulses. Each device answers only to its own address; a device in alarm reports at once, without waiting for its poll. New devices that have no address yet stay silent until the panel finds them by their serial number during auto-learn, so a loop full of new devices can be powered up without collisions.

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3 System Design Rules

Siting of devices (RDSO/SPN/217/2025 cl. 4.14)

- One ES-MSD100 multi-sensor on **either side** of each relay rack, cable termination rack, IPS equipment, power panel, change-over panel, operating panel, maintainer panel, 230 V AC point (used or unused), above each air conditioner and wherever a fire-like situation can arise.
- Adjacent detectors form a **coincidence pair**: one detector in alarm gives an alert at the panel; both give a confirmed fire.
- One ES-MCP100 call point at **each entrance / exit**, 1400 mm +/- 200 mm from the floor to the middle of the call point, visible and lit.
- DG enclosure: at least two probe-type bi-metallic heat detectors; diesel oil store: at least two UV & IR flame detectors 10 m apart - both through ES-IM100 input modules, paired for coincidence.
- Cable trays in power equipment, relay and telecom rooms: LHS cable or LHD through ES-IM100.
- Relay rooms of major telephone exchanges, data centres, MSC / BSC: aspirating detector, stages 1-4 to ES-IM100 inputs.
- Hooter cum strobe in the Station Master's room and in each protected room; it may be extended to the ESM duty room within the loop distance.

Loop planning

- At most 128 devices per loop; keep **20 % spare** addresses for expansion.
- Loop length up to 1.5 km with 1.5 sq.mm cable (loop resistance up to 50 ohm).
- Use Class A wiring (return to the panel) wherever possible.
- Keep each room on as few loops as possible, and the two detectors of a coincidence pair on the same or adjacent loop.

Cable (RDSO/SPN/217/2025 cl. 4.11)

Where	Cable
Circuit not within one building	Armoured copper, 2 core x 1.5 sq.mm min., LSZH cross-linked inner and outer sheath, 1000 V, twisted, anti-rodent
Circuit within one building	Unarmoured copper, 2 core x 1.5 sq.mm min., LSZH cross-linked sheath, 500 V, twisted, with uninsulated CPC, anti-rodent
Fire performance	950 deg C for 2 hours with water and impact, certified by LPCB or equivalent; certificate supplied
General	Fire alarm only - never share with other circuits; colour band tags; 'S' loop on both sides of each device; 2-hour fire-rated glands at junction boxes

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Battery calculation

The battery must run the system for **24 hours in standby, then 15 minutes in alarm** at full load (NFPA 72 cl. 10.6.7.2.1.2):

C (Ah) = 1.25 x [(I standby x 24 h) + (I alarm x 0.25 h)]

Table 2 **Battery calculation sheet**

Item	Quantity	Standby current (mA)	Alarm current (mA)
Control panel (4 loop cards)	1	_____	_____
ES-MSD100 multi-sensor		_____	_____
ES-MCP100 call point		_____	_____
ES-IM100 / ES-RM100 modules		_____	_____
ES-HS100 hooter cum strobe		0	_____
ES-GSM100 GSM module		_____	_____
Total			

1.25 is the ageing factor. Use the next larger standard battery size.

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4 Installing the Panel

Location

- Normally the Station Master's room, or as decided by the purchaser (cl. 4.9.5).
- Indoors, dry, 0 to 49 deg C, away from direct sun, water and heat sources.
- Display at eye level (about 1.5 m to the centre); at least 0.5 m clear in front and room for the door to open fully.
- Near a dedicated, fused mains supply with earth.

Mounting

1	Open the door and remove the cards' cover plate. Keep the screws.
2	Mark the fixing holes using the template / mechanical drawing (Annexure B). Drill and fix with four M6 screws and anchors suitable for the wall.
3	Knock out only the cable entries you need. Fit 2-hour fire-rated glands.
4	Bring the cables in, leaving 300 mm spare inside. Keep mains cables apart from loop and signal cables.
5	Earth the cabinet to the station earth.

Wiring

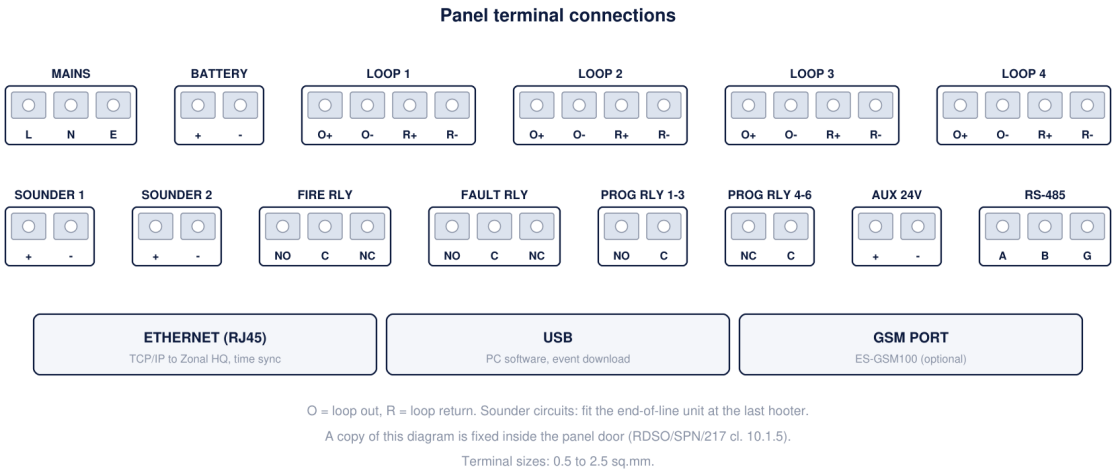


Figure 3 Panel terminals (also fixed inside the panel door)

Table 3 Terminal connections

Terminal	Connect	Notes
MAINS L N E	110 / 230 V AC, 50 Hz	Through a dedicated, labelled, lockable isolator. Fuse rating _____ A.
BATTERY + -	2 x 12 V batteries in series (24 V)	Connect last. Red to +, black to -. Reverse polarity protected.
LOOP n O+ O-	Loop out	Class A: loop returns to R+ R-. Class B: R+ R- unused.
SOUNDER 1 / 2	ES-HS100 hooter cum strobes	End-of-line unit across the last hooter.

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Terminal	Connect	Notes
FIRE RLY / FAULT RLY	Remote alarm, data logger	Potential-free, 1 A at 30 V DC. FAULT relay is energised in normal (fail-safe).
PROG RLY 1-6	Suppression, power shut-down, data logger	3 NO + 3 NC, at least 500 mA.
AUX 24V	Ancillary equipment	0.5 A maximum, protected.
RS-485 A B G	Sub-panels, PC	Twisted pair, screen to G at the panel end only.
ETHERNET	Zonal / Divisional HQ network	Through the railway network equipment.
USB	PC software	For commissioning and event download.
GSM PORT	ES-GSM100	Optional; SIM provided by the purchaser.

Loop wiring

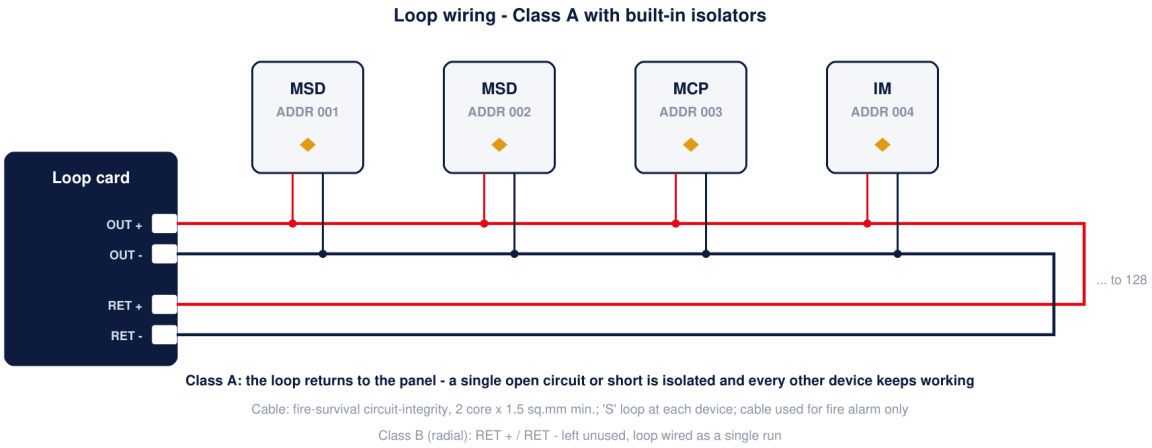


Figure 4 Loop wiring

> **!** **CAUTION** Check every loop for continuity, insulation (more than 10 M ohm at 500 V DC between cores and to earth, devices disconnected) and loop resistance (50 ohm or less) **before** connecting it to the panel.

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5 Installing the Devices

ES-MSD100 multi-sensor

1	Fix the base to the ceiling (or wall) with two screws, independent of the cables.
2	Connect L+ in / out and L- in / out. Leave an 'S' loop of cable on both sides.
3	Stick the second barcode label of the detector on the drawing / site plan at its position.
4	Twist the detector into the base until it locks. Leave the dust cover on until commissioning.

ES-MCP100 call point

Mount on the back box at 1400 mm +/- 200 mm from the floor to the middle, at entrances / exits, visible and lit. Wire L+ / L- in and out. Keep the reset key with the panel key.

ES-IM100 input and ES-RM100 relay modules

Mount in a junction box near the equipment served. Inputs: fit the end-of-line resistor supplied at the far end of each input circuit. Relay outputs: potential-free, wire as required by the suppression or power equipment; at least 500 mA per contact.

ES-HS100 hooter cum strobe

Mount at a height where it can be seen and heard (2.1 to 2.5 m). Wire in parallel on a sounder circuit with the end-of-line unit across the last one. Set the sound level (up to 90 dB at 1 m).

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6 Commissioning

1	Complete the **pre-commissioning check list** (chapter 9).
2	Switch on the mains. Then connect the battery. MAINS ON, SYSTEM ON and CHARGING light.
3	Set the date and time (**Settings > Date and time**); set the network address if the HQ link is used.
4	**Auto-learn** each loop: **Settings > Loops and devices > Auto-learn loop n**. The panel finds every device by its serial number, gives it the next free address and lists it. Each device flashes its LED three times when addressed.
5	Or enter devices by hand: **Add device**, type the serial number from the barcode label and the address.
6	Enter a **location text** (up to 32 characters) for every device, e.g. RELAY ROOM - RACK 3 LEFT.
7	Set **coincidence pairs**, **action levels**, **outputs and delays** (suppression release delay 0 to 10 min in 0.5 min steps), and **sensitivity** of each multi-sensor.
8	Enter the GSM numbers (if fitted) and send a test SMS.
9	Remove the dust covers. Test **every device** with walk test: smoke (test aerosol) and heat (heat tester) for each multi-sensor, every call point, every input. Check the location text shown is correct.
10	Test every output: hooters and strobes, relays, suppression (with the suppression system **isolated**), network, GSM. Test degrade mode by removing the CPU card while a detector is in alarm.
11	Test mains failure: switch off the mains - the panel changes to battery (STAND BY ON) within 10 seconds.
12	Print / download the event log and device list with the PC software. Hand over the system with this manual, the Operation Manual, the Do's & Don'ts card and the commissioning record.

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7 Maintenance

Table 4 Maintenance schedule

Interval	Work
Daily (user)	Panel normal; lamp test; faults recorded and reported.
Weekly (user)	One manual call point operated in turn; alarm checked.
Monthly	Hooters and strobes in every room; GSM test SMS; event log reviewed; battery voltage.
Quarterly	25 % of detectors tested with smoke and heat in turn (all within one year); dirty-chamber reports cleared; relays and outputs checked with suppression isolated.
Half-yearly	Battery: mains-off test for 30 min, terminals cleaned and tightened; earth fault check; loop resistance and insulation recorded.
Yearly	Every device tested; detectors cleaned (more often in dusty, sandy or desert areas - every _____ months, cl. 2.9); full system test including degrade mode and mains changeover; firmware and PC software versions recorded.
Every 4 years	Replace the batteries (or sooner if the half-yearly test fails).
Every 10 years	Replace the detectors.

Special tools and instruments

Table 5 Tools (cl. 6.6)

Tool	Use
Smoke detector test aerosol and dispenser pole	Smoke test of multi-sensors
Heat tester (hot-air) with pole	Heat test of multi-sensors
Detector removal tool with pole	Removing / fitting detectors
Insulation tester 500 V DC	Loop insulation (more than 10 M ohm)
Digital multimeter	Voltages, loop resistance
Laptop with SmartLoop PC software and USB cable	Configuration, event download, device values
Call point reset keys, panel keys	Reset and access

Recommended spares (cl. 4.15)

Item	Quantity
Detectors, modules, loop cards	10 % of installed quantity, at least one per installation
CPU card, power supply card	One for every five installations, at least one
Fuses, end-of-line units, call point elements	One set per installation

Spares, including bought-out items, are available for at least 10 years from the date of supply.

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8 Troubleshooting

Table 6 Troubleshooting (cl. 11.1.2.4)

Symptom	Likely cause	Remedy
LOOP OPEN CIRCUIT	Broken wire or loose terminal between the two addresses shown	Check terminals and cable between those devices; repair; RESET.
LOOP SHORT CIRCUIT	Short between the isolators shown	Find the short (insulation tester), repair, RESET.
DEVICE MISSING	Detector removed, wrong base contact, cable damage	Refit / replace the device; check wiring.
DUPLICATE ADDRESS	Two devices with the same address	Re-address one from the panel (Loops > Edit).
WRONG DEVICE TYPE	Device replaced by another type	Fit the correct type or change it in the panel.
DIRTY CHAMBER	Dust in the smoke chamber	Clean or replace the detector.
Frequent alerts from one detector	Dust, insects, steam, airflow from AC	Clean; check siting; lower sensitivity one step.
EARTH FAULT	A core touching earth	Disconnect circuits one at a time to find it; repair insulation.
MAINS FAULT	Supply off, fuse blown	Check the isolator and fuse.
BATTERY LOW / MISSING	Long mains failure, old battery, loose lead, fuse	Restore mains; check leads and fuse; test / replace.
CHARGER FAULT	Power supply card fault	Replace the power supply card.
SOUNDER CIRCUIT FAULT	Open / short in hooter wiring, EOL missing	Check wiring and end-of-line unit.
No SMS	SIM, signal, number wrong	Check SIM balance and signal; check the numbers.
HQ link down	Network cable, IP settings	Check cable and settings; ping test from the PC.
SYSTEM FAULT	CPU or internal bus fault	Replace the CPU card. Loop cards keep sounding alarms meanwhile.
Display blank, LEDs on	Display card or cable	Reseat the cable; replace the display card.

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9 Check Lists

Copy these check lists, fill them in on site and keep them with the station records (cl. 11.1.4).

Pre-installation

Check	OK
Site survey done; drawings show every device with its loop, address and location text	[ ]
Device quantities match the siting rules (cl. 4.14); 20 % spare per loop	[ ]
Panel location agreed; dedicated mains supply and earth available	[ ]
Cable as per cl. 4.11, with fire performance certificate	[ ]
Battery size calculated (24 h + 15 min)	[ ]
Cable routes, glands, junction boxes and fixings available	[ ]
Suppression system, power shut-down and data logger interfaces agreed with the purchaser	[ ]

Pre-commissioning

Check	OK
All devices fitted, bases secure, 'S' loops left, dust covers on	[ ]
Each loop: continuity, insulation > 10 M ohm, loop resistance <= 50 ohm recorded	[ ]
No loop cable shared with other circuits; colour band tags fitted	[ ]
Sounder circuits wired with end-of-line units	[ ]
Relay outputs wired; suppression system ISOLATED for testing	[ ]
Mains and earth checked; battery voltage and polarity checked	[ ]
Connection diagram fixed inside the panel door	[ ]

Maintenance visit

Check	OK
Panel normal; event log reviewed and downloaded	[ ]
Lamp test; buzzer; display	[ ]
Devices tested as per schedule (list addresses tested)	[ ]
Hooters and strobes; relays; GSM SMS; HQ link	[ ]
Battery mains-off test; charger voltage	[ ]
Detectors cleaned as required; dirty-chamber reports cleared	[ ]
System returned to normal; nothing left disabled; register signed	[ ]

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10 Do's and Don'ts

Print this page on card, laminate it and keep it at the panel (cl. 11.1.3).

DO	DON'T
Read the loop, address and location on the display in alarm	Don't silence or reset before checking the location
Do a lamp test every day	Don't leave the key in the key switch
Report every fault at once	Don't disable a loop or device without authority
Enable everything again after work	Don't cover, paint or tape detectors
Keep detectors clean; clean more often in dusty areas	Don't store material within 0.5 m of a detector
Keep call points visible, lit and unobstructed	Don't use the fire alarm cable for other circuits
Isolate the suppression system before testing	Don't open the panel with mains on
Record every event in the fire register	Don't fit devices from other makes on the loop

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11 Hardware Details and Drawings

Schematic diagrams and part lists of every assembled PCB (cl. 11.1.2.2) and mechanical drawings of every unit (cl. 11.1.2.3) are supplied with each system as the annexures below.

Annexure	Document	Drawing / document no.	Rev.
A1	CPU card - schematic and part list	_____	
A2	Loop card - schematic and part list	_____	
A3	Display & keypad card - schematic and part list	_____	
A4	I/O card - schematic and part list	_____	
A5	Comms card - schematic and part list	_____	
A6	Power supply card - schematic and part list	_____	
A7	ES-MSD100, ES-MCP100, ES-IM100, ES-RM100 - schematics and part lists	_____	
B1	Control panel - mechanical drawing and fixing template	_____	
B2	Field devices - mechanical drawings	_____	
C1	Typical station layout and cable schedule	_____	

Marking

Each unit is marked with the manufacturer's name and address, year of manufacture, serial number and specification number RDSO/SPN/217/2025 Version 3.1; the panel carries the connection diagram inside its door (cl. 10.1).

Warranty and training

Warranty: 24 months from the date of supply (cl. 12). Infusive Indotech Private Limited provides training in operation, maintenance, inspection and testing (cl. 13), supervises installation and takes part in commissioning, testing and field trials (cl. 6.4, 6.5).

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EMBER SAFE

STAY ALERT

SmartLoop-4 - 4 Loop Analogue Addressable Fire Alarm Control Panel

Designed and manufactured in India

INFUSIVE INDOTECH PRIVATE LIMITED

Address: _____

Phone: _____ E-mail: _____

Website: _____

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